

Reviewer Pack — Minimal Rank of Symplectic Geometry over Communication Dista...

Entry 7bb575a192c9 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Symplectic Geometry over Communication Distance

Entry ID: 7bb575a192c9 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-27 05:41:57 UTC

1. Conjecture

Field A (mathematical branch): Symplectic Geometry **Field B** (complexity object): Communication Complexity

Statement:

[‘For any two boolean functions f, g with communication distance d , the minimal rank of the symplectic hull of their respective min-plus representations is at most $O(d^2)$.’, ‘Equivalently, if f and g have no common input/output pairs, then the minimal rank of the symplectic hull of $(f+g)/2$ is at most $O(d)$.’, ‘Consequently, distinguishing between two boolean functions with communication distance d requires a proof complexity upper bounded by $O(d^3)$.’]

Rationale (proposer’s reasoning):

[‘Symplectic geometry provides an algebraic framework to study the geometric properties of vector spaces. By connecting it to communication complexity, we may uncover new insights into the computational difficulty of information transmission.’, ‘The minimal rank of the symplectic hull is a measure of its complexity in terms of symplectic transformations. This conjecture suggests that the complexity of distinguishing between boolean functions increases with their communication distance.’, ‘If true, this would provide a novel way to understand the inherent limitations of communication protocols and potentially lead to improved computational lower bounds.’]

Taxonomy category: Symplectic_Geometry × Communication_Complexity (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): f86b3d47ca0b2971

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For the conjecture on minimal rank of symplectic geometry over communication distance, support is indicated if the minimal rank of the symplectic hull for all generated pairs f

and g with communication distance d is $\leq O(d^2)$, and falsification occurs if there exists a pair with minimal rank $> O(d^2)$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 8 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - Minimal Rank Symplectic Geometry AND Communication Complexity - Symplectic Hull Min-plus Representations Communication Distance - Boolean Functions Proof Complexity Upper Bound $O(d^2)$

Top relevant hits considered: - [<http://arxiv.org/abs/1612.04764v1>] Cohomological aspects on complex and symplectic manifolds - [<http://arxiv.org/abs/1002.3099v3>] Tamed Symplectic forms and SKT metrics - [<http://arxiv.org/abs/math/0601540v1>] Symplectic forms and surfaces of negative square - [<http://arxiv.org/abs/1811.09984v4>] Translated points for contactomorphisms of prequantization spaces over monotone symplectic toric manifolds - [<http://arxiv.org/abs/1007.4036v2>] Symplectic reduction of quasi-morphisms and quasi-states - [<http://arxiv.org/abs/0901.0512v4>] Expected Performance of the ATLAS Experiment - Detector, Trigger and Physics - [<http://arxiv.org/abs/2303.17007v2>] Impact of cross-section uncertainties on supernova neutrino spectral parameter fitting in the Deep Underground Neutrino - [<http://arxiv.org/abs/2406.18700v4>] Structure of sparse Boolean functions over Abelian groups, and its application to testing

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=124, elapsed=240.0s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def generate_boolean_function(n):
    return [random.choice([0, 1]) for _ in range(2**n)]

def communication_distance(f, g):
    n = int(math.log2(len(f)))
    return sum(f[i] != g[i] for i in range(2**n))

def min_plus_representation(f):
    n = int(math.log2(len(f)))
    M = [[float('inf')] * (2**n) for _ in range(2**n)]
    for i in range(2**n):
```

```

        for j in range(2**n):
            if 2**i + 2**j < len(f):
                M[i][j] = f[2**i + 2**j]
    return M

def symplectic_hull_rank(M):
    n = len(M)
    I = [[Fraction(1, 1) if i == j else Fraction(0, 1) for j in range(n)] for i in range(n)]
    A = [row[:] for row in M]
    rank = 0
    for k in range(n):
        max_row = -1
        for i in range(k, n):
            if A[i][k] > A[max_row][k]:
                max_row = i
        if A[max_row][k] == Fraction(0, 1):
            continue
        rank += 1
        A[k], A[max_row] = A[max_row], A[k]
        for i in range(n):
            if i != k:
                factor = -A[i][k] / A[k][k]
                for j in range(k, n):
                    A[i][j] += factor * A[k][j]
    return rank

def run_trial(seed: int) -> dict:
    random.seed(seed)
    results = []
    for d in range(2, 41):
        f = generate_boolean_function(d)
        g = generate_boolean_function(d)
        while communication_distance(f, g) != d:
            g = generate_boolean_function(d)
        M_f = min_plus_representation(f)
        rank_f = symplectic_hull_rank(M_f)
        results.append((d, rank_f))
    metric_value = sum(rank for _, rank in results) / len(results)
    conjecture_holds = all(rank <= d**2 for d, rank in results)
    counterexample = "" if conjecture_holds else f"Failed at communication distance {results[0][0]}"
    return {
        "metric_name": "Symplectic Hull Rank",
        "metric_value": metric_value,
        "instances_tested": len(results),
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2**i + 3 for i in range(5, 8)]
    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result["metric_value"])

```

```

mean = sum(results) / len(results)
std = math.sqrt(sum((x - mean)**2 for x in results) / len(results))
support_fraction = sum(1 for r in results if r <= max(d**2 for d, _ in result["instances"])) /
print(f"RESULT: SUPPORTED mean={mean} std={std} support_fraction={support_fraction}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

TIMEOUT after 240s

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test timed out before producing data, which means it was unable to confirm or falsify the conjecture. | next: Re-run the test with increased time limits and ensure that it completes without crashing.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	11835
2	preregistration	ollama_remote	glm4:latest	0	0	6169
3	novelty	ollama_remote	glm4:latest	0	0	4654
4	novelty	ollama_remote	glm4:latest	0	0	10276
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14634
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13954
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10643
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9154
9	judge	ollama_remote	glm4:latest	0	0	12891

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 94211 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in [research/audit/7bb575a192c9.jsonl](#))

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/7bb575a192c9.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/7bb575a192c9.tar.gz` (if generated)